4 **(a)** Define the electric potential at a point in an electric field.

the work done per unit positive charge in

………………………………………………………………………………………………

bringing a small test charge from infinity to that point

……………………………………………………………………………………… [1]

(b) Fig. 4.1 shows part of the region between two charges of the same magnitude but opposite sign.

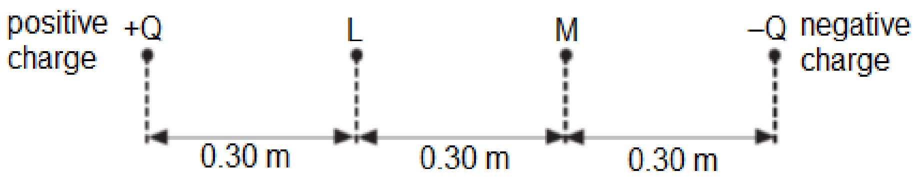


Fig. 4.1

(i) The electric potential at point **L** due to the positive charge **only** is +3.0 V.

Calculate the magnitude Q of the positive charge.

$V = \dfrac{Q}{4\pi\varepsilon_0 r}$ (given on page 3)

$3.0 \times (4\pi\varepsilon_0)(0.30) = Q$

$Q = 1.00 \times 10^{-10}$ C

Q = ………………………… nC [1]

(ii) Hence or otherwise, calculate the **net** electric potential at point **L**.

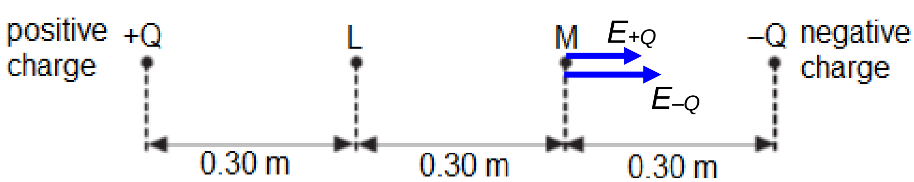
potential due to –Q = –3.0/2 V (double the distance, halve the potential) as $V \propto 1/r$

Sum = +3.0 V + (–3.0/2) = 1.5 V

** potential is a scalar while field strength is a vector. Disregarding direction, simply sum the potentials. Negative charges give negative potentials.

net electric potential = ………………………… V [2]

(iii) Calculate the **resultant** electric field strength at point **M**.



Sum of both

$$E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

$$= \frac{1.00\times10^{-10}}{4\pi(8.85\times10^{-12})(0.60)^2} + \frac{1.00\times10^{-10}}{4\pi(8.85\times10^{-12})(0.30)^2}$$

$= 12$ V m^{-1}

ecf from **4(b)(i)**

electric field strength at **M** = ………………………….. V m^{-1} [2]

(c) R and S are two charged parallel plates, 0.60 m apart, as shown in Fig. 4.2. They are at potentials of + 3.0 V and + 1.0 V respectively.

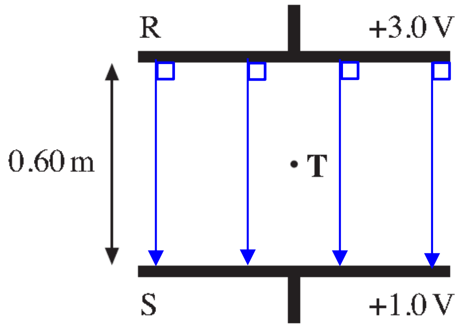


Fig. 4.2

(i) On Fig. 4.2, sketch the electric field between R and S, showing its direction.
uniformly spaced out down arrows [1]

**uniformly spaced out – constant E field between parallel plates contrasting with radial field in previous parts.
**down arrows – direction of E is the same as direction of electric force on a POSITIVE test charge.
**just need minimum 3 lines to show uniform spacing

(ii) Point **T** is mid-way between R and S.

Calculate the electric field strength at **T**.

$$E = \frac{\Delta V}{d} = \frac{3.0 - 1.0}{0.60}$$
$$= 3.3 \text{ V m}^{-1}$$

electric field strength at **T** = ………………………… V m^{-1} [2]

(iii) A point charge is suspended at point T.
State the sign of the charge and sketch an arrow on Fig. 4.2 showing the direction of the electric force.

** downwards weight = upwards electric force
** since electric force is upwards, electric field strength E (defined for postive charge) is downwards, hence charge is negative.

negative and upwards
sign of charge = …………………………….. [1]

5	**(a)**	Define a magnetic field.